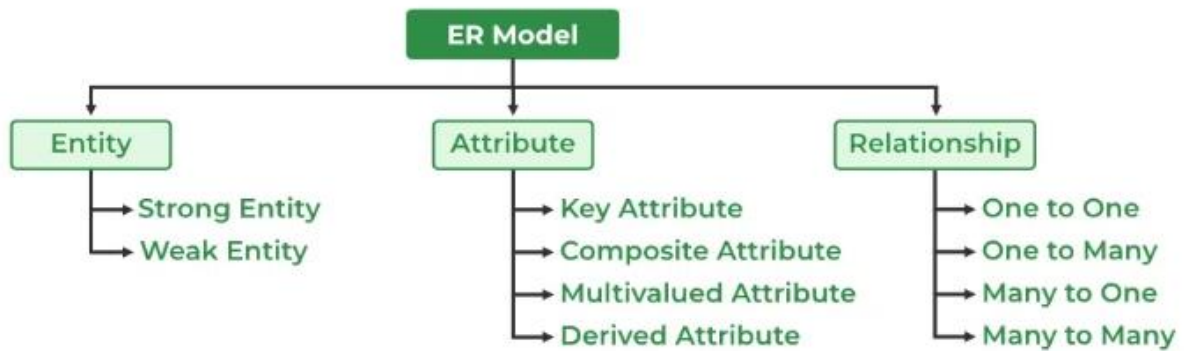


Database Design and ER Diagrams



1. Database Design

Definition

Database Design is the process of creating a database structure to store, manage, and retrieve data efficiently.

It helps:

- Organize data properly
- Reduce duplication
- Improve data security
- Increase performance

2. Objectives of Database Design

- Store data efficiently
 - Avoid data redundancy
 - Maintain data consistency
 - Provide easy data access
 - Improve security
 - Support future modifications
-

3. Steps in Database Design

Step 1: Requirement Collection

Collect information from users about:

- What data is needed
- What operations are performed
- Reports required

Example

In a college database:

- Student details
 - Faculty details
 - Course information
 - Marks information
-

Step 2: Conceptual Design

High-level design using ER Diagram.

Focuses on:

- Entities
 - Attributes
 - Relationships
-

Step 3: Logical Design

Convert ER model into tables.

Includes:

- Primary Keys
- Foreign Keys
- Constraints

Step 4: Physical Design

Actual implementation in DBMS.

Includes:

- Storage methods
 - Indexing
 - File organization
-

4. ER Diagram (Entity Relationship Diagram)

Definition

ER Diagram is a graphical representation of entities, attributes, and relationships in a database.

Used to:

- Visualize database structure
 - Simplify database planning
 - Improve understanding
-

5. Components of ER Diagram

A) Entity

Definition

An entity is a real-world object about which data is stored.

Examples

- Student
- Employee
- Course
- Book

Symbol

Rectangle

Example:

```
+-----+  
| STUDENT |  
+-----+
```

B) Attribute

Definition

Attributes describe properties of an entity.

Examples

For STUDENT:

- Student_ID
- Name
- Age
- Phone

Symbol

Oval

Example:

(Student_Name)

Types of Attributes

1. Simple Attribute

Cannot be divided further.

Example:

- Age
 - Gender
-

2. Composite Attribute

Can be divided into smaller parts.

Example:

- Address → City, State, PIN
-

3. Single-Valued Attribute

Contains only one value.

Example:

- Roll Number
-

4. Multi-Valued Attribute

Contains multiple values.

Example:

- Phone Numbers
-

5. Derived Attribute

Calculated from another attribute.

Example:

- Age derived from Date of Birth

6. Key Attributes

Primary Key

Uniquely identifies each record.

Example:

- Student_ID

Properties

- Unique
 - Cannot be NULL
-

7. Relationship

Definition

Relationship shows association between entities.

Examples

- Student ENROLLS in Course
- Teacher TEACHES Subject

Symbol

Diamond

Example:

STUDENT ---- ENROLLS ---- COURSE

8. Types of Relationships

1. One-to-One (1:1)

One entity related to one entity.

Example

- One person → One passport

2. One-to-Many (1:M)

One entity related to many entities.

Example

- One department → Many employees

3. Many-to-Many (M:N)

Many entities related to many entities.

Example

- Students ↔ Courses

9. Cardinality

Definition

Cardinality specifies the number of entities participating in a relationship.

Types

- 1:1
- 1:M
- M:N

10. Participation Constraints

Total Participation

All entities must participate.

Partial Participation

Some entities participate.

11. Weak Entity

Definition

Entity that depends on another entity for existence.

Example

- Dependent of Employee

Features

- No primary key of its own
 - Depends on strong entity
-

12. Strong Entity

Definition

Entity that has its own primary key.

Example

- Student
 - Employee
-

13. ER Diagram Symbols

Symbol	Meaning
Rectangle	Entity
Oval	Attribute
Diamond	Relationship
Double Rectangle	Weak Entity
Double Oval	Multivalued Attribute

Underlined Attribute Primary Key

14. Example ER Diagram

College Database Example

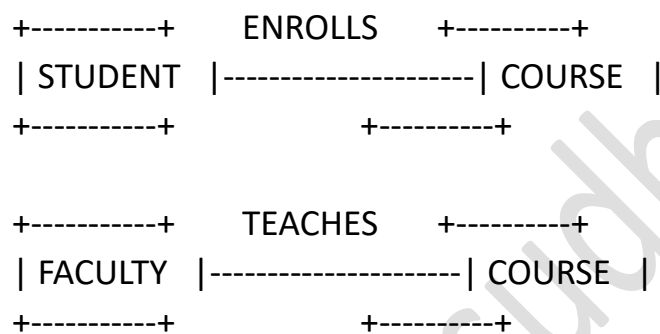
Entities:

- STUDENT
- COURSE
- FACULTY

Relationships:

- STUDENT enrolls in COURSE
- FACULTY teaches COURSE

Simple Representation:



15. Advantages of ER Diagrams

- Easy to understand
- Helps in database planning
- Reduces design errors
- Improves communication
- Provides clear structure

16. Disadvantages of ER Diagrams

- Difficult for very large databases
- Requires proper understanding

- Complex relationships may confuse users

17. Difference Between Entity and Attribute

Entity	Attribute
Real-world object	Property of entity
Example: Student	Example: Name
Stored as table	Stored as column

18. Difference Between Strong Entity and Weak Entity

Strong Entity	Weak Entity
Has primary key	No primary key
Independent	Depends on another entity
Example: Employee	Example: Dependent

19. Applications of ER Diagrams

- Banking systems
- College management systems
- Library systems
- Hospital management
- E-commerce applications

20. Conclusion

- Database Design helps organize data efficiently.
- ER Diagram is used to model database structure visually.
- Main components are:

- Entity
 - Attribute
 - Relationship
- Good database design improves performance and reduces redundancy.

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